Continuous embeddings of finite multi-seminormed spaces: an exact separatedness dichotomy

Candidate partial result for arXiv:2103.11049, Problem 5.3

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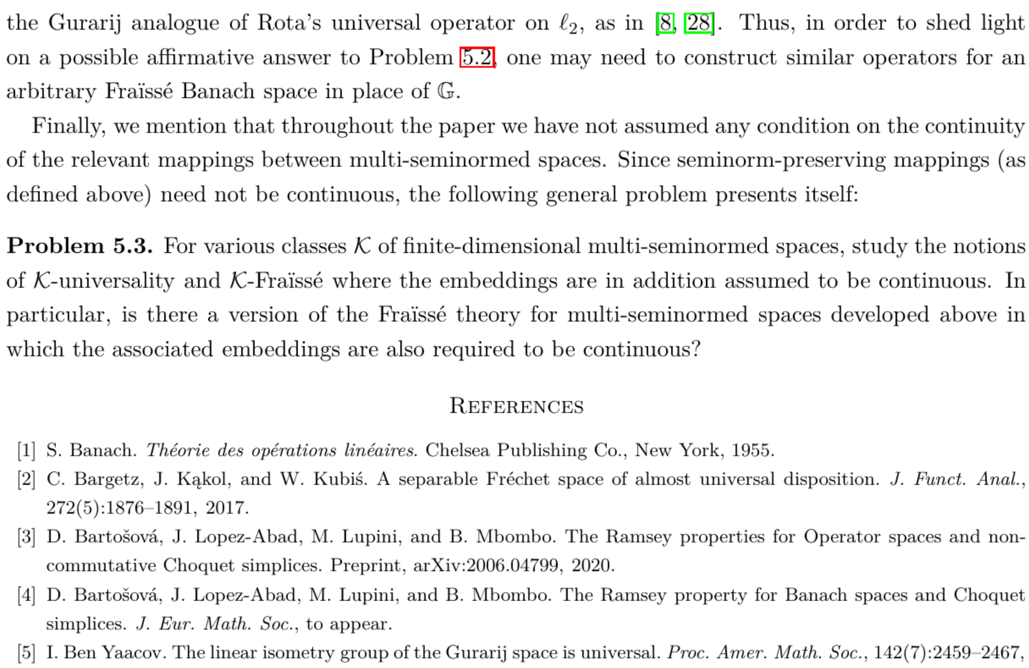
Status. Candidate partial result, likely valid. The result gives a sharp affirmative version of the final existence question in Problem 5.3 and a matching obstruction. The problem's broader request to classify particular classes and their limits is not claimed solved.

Source question

Kawach and López-Abad ask in Problem 5.3 of [1] for notions of universality and Fraïssé theory in which embeddings of finite-dimensional multi-seminormed spaces are also required to be continuous.


the Gurarij analogue of Rota's universal operator on ℓ_2, as in [8, 28]. Thus, in order to shed light on a possible affirmative answer to Problem 5.2, one may need to construct similar operators for an arbitrary Fraïssé Banach space in place of $\mathbb{G}$.

Finally, we mention that throughout the paper we have not assumed any condition on the continuity of the relevant mappings between multi-seminormed spaces. Since seminorm-preserving mappings (as defined above) need not be continuous, the following general problem presents itself:

Problem 5.3. For various classes $\mathcal{K}$ of finite-dimensional multi-seminormed spaces, study the notions of $\mathcal{K}$-universality and $\mathcal{K}$-Fraïssé where the embeddings are in addition assumed to be continuous. In particular, is there a version of the Fraïssé theory for multi-seminormed spaces developed above in which the associated embeddings are also required to be continuous?

References

[1] S. Banach. *Théorie des opérations linéaires.* Chelsea Publishing Co., New York, 1955.

[2] C. Bargetz, J. Kąkol, and W. Kubiś. A separable Fréchet space of almost universal disposition. *J. Funct. Anal.*, 272(5):1876–1891, 2017.

[3] D. Bartošová, J. Lopez-Abad, M. Lupini, and B. Mbombo. The Ramsey properties for Operator spaces and non-commutative Choquet simplices. Preprint, arXiv:2006.04799, 2020.

[4] D. Bartošová, J. Lopez-Abad, M. Lupini, and B. Mbombo. The Ramsey property for Banach spaces and Choquet simplices. *J. Eur. Math. Soc.*, to appear.

[5] I. Ben Yaacov. The linear isometry group of the Gurarij space is universal. *Proc. Amer. Math. Soc.*, 142(7):2459–2467,

The screenshot is from page 25 of the source PDF. In the terminology below, a multi-seminormed space is *separated* when the intersection of the kernels of its defining seminorms is zero.

Proof intuition

The obstruction to continuity is exactly the invisible subspace on which all source seminorms vanish. It is the closure of zero in the source topology. A continuous map must send it into the corresponding invisible subspace of the target. Once this condition holds, the map factors topologically through the finite-dimensional Hausdorff quotient, where every seminorm is bounded.

Thus continuity is automatic on separated finite-dimensional sources, whereas a non-separated source cannot inject continuously into any Hausdorff Fréchet space. This is precisely the boundary needed in the source paper: after restricting to separated finite objects, its pushouts and all their structure maps remain valid and are automatically continuous. The restriction also repairs the hereditary-property failure noted by the authors.

The exact continuity criterion

Let

$$X = (X, (p_i)_{i<\ell}), \qquad Y = (Y, (q_j)_{j<m})$$

be multi-seminormed spaces, where X is finite-dimensional and m may be finite or countable. Put

$$\mathcal{N}_X = \bigcap_{i<\ell} \ker p_i, \qquad \mathcal{N}_Y = \bigcap_{j<m} \ker q_j.$$

Theorem 1 (finite-source continuity criterion)**.** *A linear map $T : X \to Y$ is continuous for the locally convex topologies generated by the displayed seminorms if and only if*

$$T(\mathcal{N}_X) \subseteq \mathcal{N}_Y.$$

Consequently:

1. *if X is separated, every linear map $X \to Y$ is continuous;*
2. *if Y is separated and T is both continuous and injective, then X is separated.*

Proof. The closure of zero in X is exactly $\mathcal{N}_X$, and likewise $\overline{\{0\}}^{,Y} = \mathcal{N}_Y$. Continuity therefore implies $T(\mathcal{N}_X) \subseteq \mathcal{N}_Y$.

Conversely, suppose this inclusion holds. Set

$$r(x) = \max_{i<\ell} p_i(x).$$

This is a seminorm with kernel $\mathcal{N}_X$, and it induces a norm $\bar{r}$ on the finite-dimensional quotient $X/\mathcal{N}_X$. For each $j < m$, the seminorm $q_j \circ T$ vanishes on $\mathcal{N}_X$, hence descends to a seminorm s_j on $X/\mathcal{N}_X$. Since that quotient is finite-dimensional, there is $C_j < \infty$ such that

$$q_j(Tx) = s_j(x + \mathcal{N}_X) \le C_j \bar{r}(x + \mathcal{N}_X) = C_j r(x) \qquad (x \in X).$$

This is the continuity estimate for every target seminorm q_j, so T is continuous. If X is separated, then $\mathcal{N}_X = 0$ and the criterion is automatic. If Y is separated and T is continuous, then $T(\mathcal{N}_X) = 0$; injectivity gives $\mathcal{N}_X = 0$. □

The maximal continuous Fraïssé domain

Theorem 2 (sharp Hausdorff dichotomy)**.** *Let E be a Hausdorff Fréchet space and let $\mathcal{K}$ be a class of finite-dimensional multi-seminormed spaces.*

1. *If E is $\mathcal{K}$-universal via continuous injective linear maps, then every member of $\mathcal{K}$ is separated.*
2. *On any class consisting of separated finite-dimensional objects, adding continuity to the paper's linear multi-isometric or multi-δ-isometric embedding requirement does not remove a single morphism.*

Hence the separated finite-dimensional subcategory is the maximal possible domain for a continuous-embedding theory with Hausdorff Fréchet limits.

Proof. For (1), apply Theorem 1(2) to every continuous injective map $X \to E$, using separatedness of E. Part (2) is Theorem 1(1). □

There is also a positive Fraïssé consequence, not merely an obstruction.

Corollary 3 (a continuous-morphism Fraïssé theory)**.** *Use continuous multi-isometric embeddings as morphisms. The classes of all separated finite-dimensional multi-seminormed spaces of any fixed finite length, of length at most a fixed finite integer, or of arbitrary finite length, are amalgamation classes with stability modulus*

$$\varpi(d, \ell, \delta) = 2\delta.$$

Relative to the continuous-morphism category they also have the hereditary property, and the source paper's Fraïssé construction produces their Hausdorff Fréchet limits. The same conclusions hold for the corresponding graded classes.

Proof. Between separated finite-dimensional objects all linear maps are continuous by Theorem 1. Therefore the pushout in Proposition 2.3 of [1] is unchanged: its amalgam is separated, its structure maps are continuous, and the proof gives the same bound 2δ.

The source notes that its separated classes fail the hereditary property under the unrestricted morphisms because a non-separated object can embed into a separated one. Under continuous morphisms this is impossible by Theorem 1(2), so heredity is restored. The metric separability and Banach–Mazur closure arguments are unchanged because the morphisms between separated objects have not changed. The Fraïssé-limit construction of Section 3 of [1] now applies verbatim. Its finite-dimensional structure maps are continuous by Theorem 1(1), and its separated completion is Hausdorff. The graded pushout in the same proposition gives the final assertion. □

Scope, verification, and novelty

Theorem 2 rules out retaining the non-separated finite truncations of the paper's full classes $\mathcal{M}_{<\omega}$ and $\mathcal{G}_{<\omega}$ when both continuous injective embeddings and a Hausdorff Fréchet limit are required. Corollary 3 gives the maximal direct replacement. It does not identify all resulting limits or settle the open-ended request to study every special class $\mathcal{K}$.

The proof was checked seminorm by seminorm; it uses no computation. A bounded novelty search on 11 August 2026 covered the local run indexes and exact-phrase and keyword web searches for Problem 5.3 and continuous multi-seminormed Fraïssé embeddings. No later answer was found. Novelty confidence is moderate because the finite-dimensional continuity criterion may be folklore; its sharp application to this problem was not located.

Human-review recommendation. Verify that the intended continuous category formulates heredity only with continuous morphisms. Under that natural reading, the topological dichotomy is exact and the transfer of the source pushout is immediate.

References

[1] J. K. Kawach and J. López-Abad, *Fraïssé and Ramsey properties of Fréchet spaces*, J. Math. Anal. Appl. 507 (2022), 125769; arXiv:2103.11049.